

Advanced Light Intensity Indicator (ALII)

Digital Signal Processing Hardware Project Concise Technical Summary

Project Type	Analog and digital discrete-hardware design
Primary Sensor	Light Dependent Resistor (LDR)
Digital Resolution	3-bit light level, displayed as 0–7
Implementation Constraint	No programmable ICs or task-specific pre-built ICs

Team Members

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1 Project Overview

The Advanced Light Intensity Indicator (ALII) is a hardware system developed to sense, condition, digitize, and display ambient light intensity. In addition to showing the current light level, the system includes a stability-detection function to reject short-term fluctuations and a time-averaging function to provide a representative light level over a longer interval. The complete design is implemented using analog components and discrete digital logic rather than a microcontroller.

2 Original Design Requirements

- Use an LDR to sense ambient light intensity.
- Include an analog filter before processing to suppress unwanted electrical interference, especially 50–100 Hz power-line-related noise.
- Display light intensity as one of eight levels, 0 through 7, on a seven-segment display.
- Provide stability detection so the displayed stable value changes only after the light remains stable for an adjustable 30–300 s period, with an enable/disable switch.
- Provide a second processed output showing average light intensity over an adjustable 300–900 s period.
- Include a reset control for the averaging function.
- Do not use programmable ICs or pre-built ICs that directly perform the required task; gate ICs and flip-flops are allowed.

3 Implemented Architecture

```
Ambient Light -> LDR -> Analog Low-Pass Filter -> Flash ADC -> 3-bit Level (0-7)
|-> Instantaneous Display
|-> Stability Detection -> Stable Display
|-> Time Averaging -> Average Display
```

4 Functional Blocks

4.1 LDR Light-Sensing Stage

The LDR produces an analog signal that varies with ambient light. This is the raw measurement signal used by the rest of the system.

4.2 Analog Signal Conditioning

A low-pass filtering stage is placed before the ADC section. Its purpose is to reduce power-line interference and harmonics that could otherwise cause flickering or unstable digital readings. The filter therefore improves measurement reliability while preserving genuine changes in ambient illumination.

4.3 Flash ADC and 3-bit Quantization

The conditioned analog signal is converted into a digital light level using a Flash ADC architecture. A resistor ladder generates multiple reference voltages and comparators determine the input range. The resulting information is encoded as a 3-bit value, giving eight possible levels from 000 to 111, corresponding to display values 0 to 7.

3-bit Code	Light Level
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

4.4 Instantaneous Display

The digitized 3-bit light level is decoded and displayed on a seven-segment display. This gives an immediate visual indication of the current light intensity from 0 (lowest) to 7 (highest).

4.5 Stability Detection

The stability section prevents temporary light fluctuations from immediately changing the processed output. The target stability time is adjustable from 30 to 300 seconds. Based on the implementation description, the function was realized using timing hardware together with relays/logic and flip-flops. The stable display updates only when the measured light level remains unchanged for the selected duration.

4.6 Time-Based Averaging

The averaging section determines the representative light intensity over a user-adjustable interval of 300 to 900 seconds. The implementation uses digital counters, adders, and flip-flops to accumulate/process the light-level information over time. A reset input restarts the averaging process when required.

5 Key Design Decisions

- Filtering is performed before digitization so electrical noise does not propagate into the comparator and logic stages.
- A Flash ADC is suitable because only eight light levels are needed; 3-bit resolution is sufficient for values 0–7.
- Discrete sequential logic is used to satisfy the restriction against programmable controllers.
- Separate instantaneous, stable, and average outputs allow the system to show both immediate conditions and processed long-term information.
- Adjustable timing makes the design adaptable to different lighting environments and demonstration conditions.

6 Validation / Demonstration Checklist

1. Change the simulated LDR/light-brightness input and verify that the instantaneous display changes through valid levels 0–7.
2. Introduce or observe high-frequency/power-line-related disturbance and verify that the conditioned signal is smoother than the raw sensor signal.
3. Change the light level briefly and verify that the stable output does not update before the selected stability period.
4. Keep the light level unchanged for the selected stability period and verify that the stable display

updates.

5. Enable and disable the stability feature and verify the expected change in behavior.
6. Run the averaging section over the selected averaging period and verify that the average display represents the accumulated light level.
7. Use the averaging reset control and confirm that the averaging process restarts.

7 Learning Outcomes

- Practical use of breadboard, analog, and digital hardware components.
- Design and analysis of low-pass signal-conditioning circuits.
- Understanding of Flash ADC operation, resistor ladders, comparators, and 3-bit quantization.
- Implementation of real-world logic using combinational and sequential circuits.
- Use of counters, adders, flip-flops, timing circuits, and display logic.
- Hardware and simulation troubleshooting and subsystem integration.

8 Team Members

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9 Recommended GitHub Repository Structure

```
Advanced-Light-Intensity-Indicator/  
|-- README.md  
|-- docs/  
|   |-- ALII_Project_Summary.pdf  
|   '-- Project_Requirements.pdf  
|-- simulation/  
|   '-- ALII_Final_Simulation.txt  
'-- images/  
    |-- final_breadboard.jpg  
    |-- full_simulation.png  
    |-- low_pass_filter.png  
    |-- flash_adc.png  
    |-- stability_detection.png  
    |-- time_averaging.png  
    '-- seven_segment_display.png
```

10 Conclusion

The ALII project demonstrates how a relatively complex signal-processing task can be implemented using discrete analog and digital hardware. The system combines sensing, filtering, Flash ADC conversion, digital encoding, timing, sequential logic, averaging, and seven-segment display functions while satisfying the restriction against programmable ICs. It provides a clear practical example of integrating analog signal conditioning with digital signal-processing concepts in hardware.